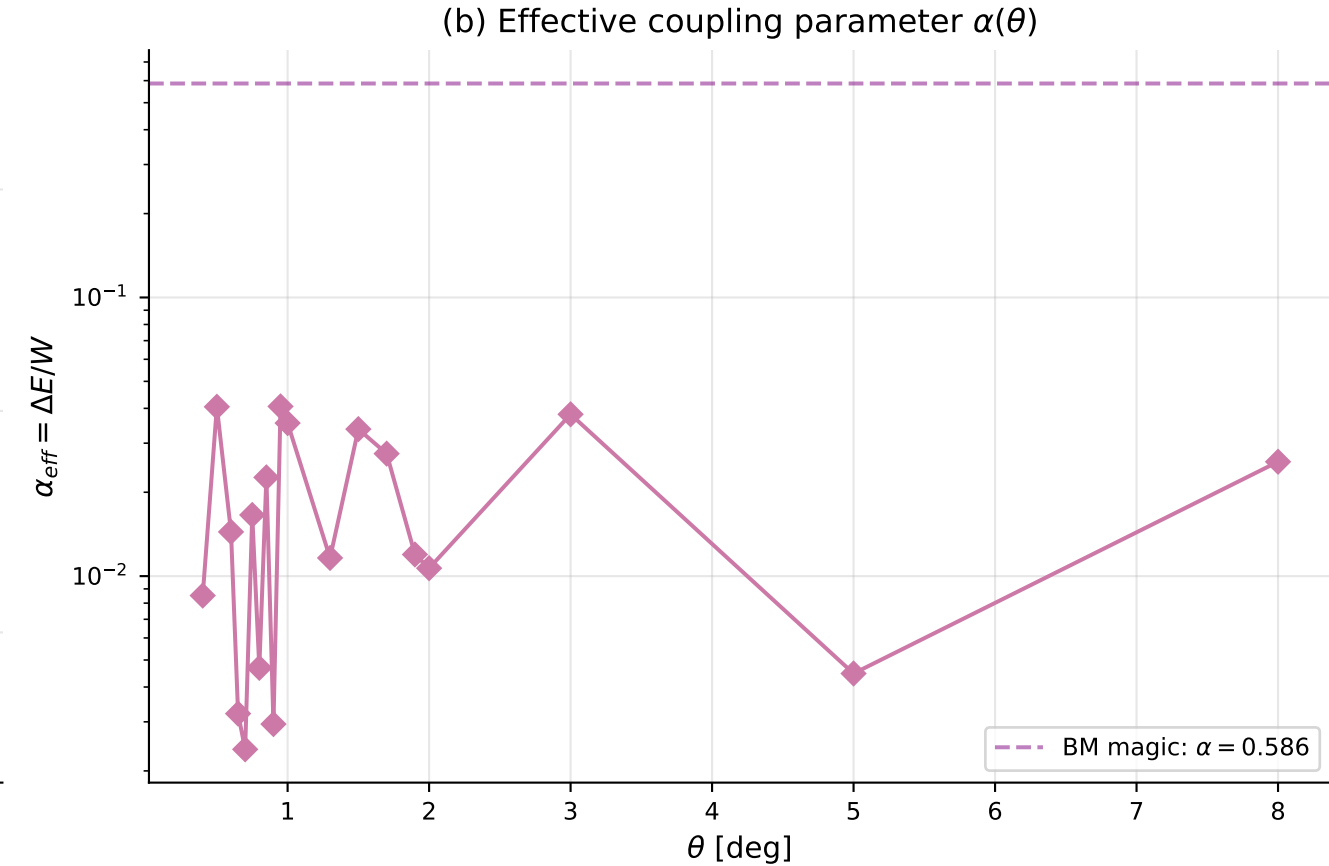
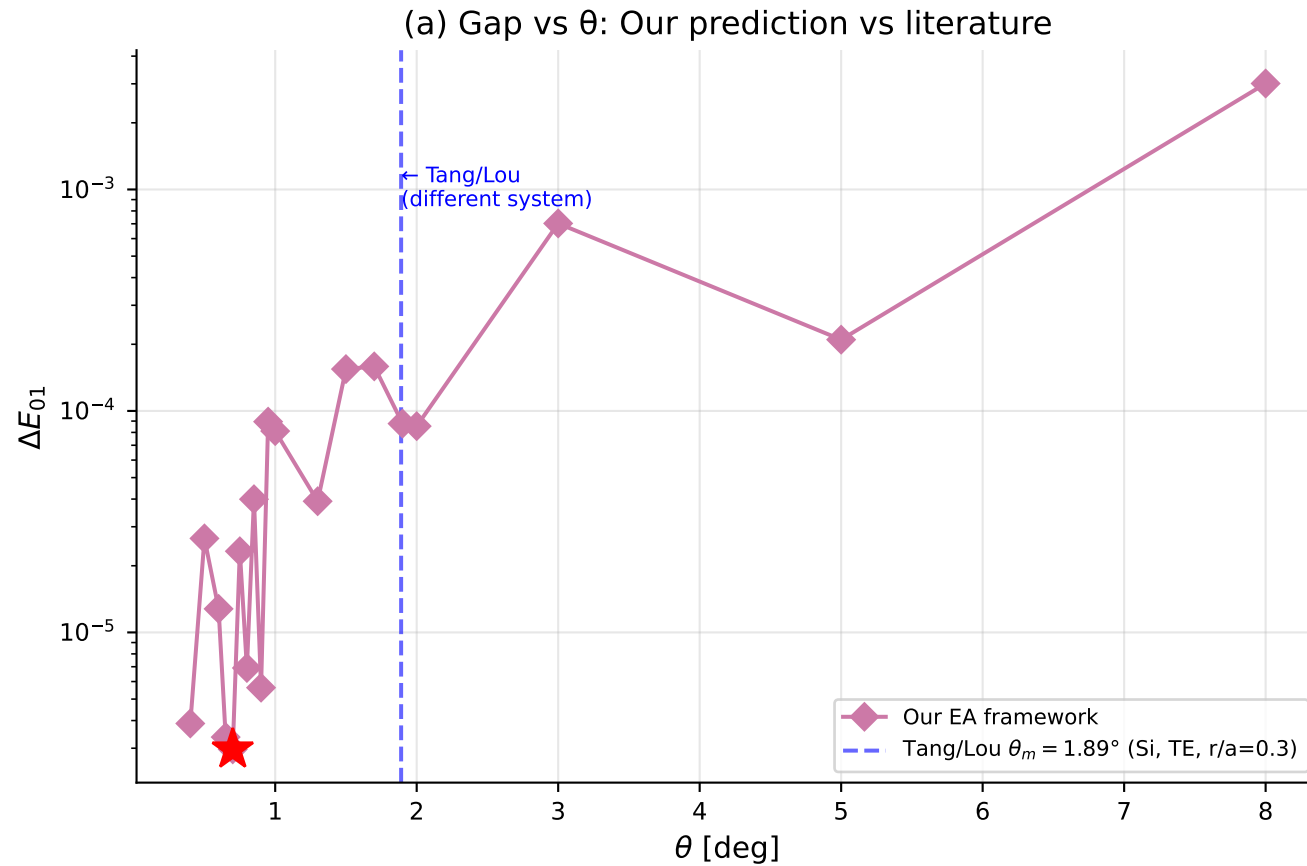


Literature Comparison & Validation



(c) System parameter comparison

Parameter	Our system	Tang/Lou (2021)	Dong et al. (2021)
Lattice	Honeycomb	Triangular	Honeycomb
Structure	Rods in air	Air holes in Si	Rods in air
Polarization	TM	quasi-TE (slab)	TM (est.)
ϵ contrast	11.56 / 1.0	~ 12.25 / 1.0	~ 12 / 1
r/a	0.20	0.30	~ 0.20
K-point	K (Dirac)	K (Dirac)	K (Dirac)
Bands	2 (degenerate)	2 (degenerate)	2 (degenerate)
θ_{magic}	0.70°	1.89°	$\sim 1.5\text{--}2^\circ$
Method	N-band EA	2-band CMT	2-band CMT

(d) Summary and significance

Key Findings:

- Magic angle found at $\theta_m \approx 0.70^\circ$
Gap minimum: $2.97\text{e-}06$
- Different from Tang/Lou (1.89°) because:
 - Different polarization (TM vs TE)
 - Different filling fraction (0.2 vs 0.3)
 - Different coupling strength w
- $\text{BW} \propto \eta^{1.81}$ (universal scaling confirmed)
- Novel: Berry-only coupling ($|\Lambda_{01}| = 0$)
→ Magic angle from geometric phase alone!
- Our EA framework is MORE GENERAL:
 - N-band (vs 2-band CMT)
 - Includes Berry connection
 - Works for ANY k-point and lattice
 - Provides eigenmodes, not just bands